Sahaj Bhadja

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Summary

Software engineer with 2+ years building **distributed backend systems**, **secure cloud workflows**, and **production APIs**. Strong in **TypeScript**-adjacent platform engineering, reliability, and fault-tolerant execution for blockchain-oriented infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected **multi-tenant Azure SQL and Blob Storage services** for 500+ users, tuning indexed queries and backend access paths to reduce dashboard latency from **1.2 seconds to 350 milliseconds** across production workflows.
- Built **fault-tolerant background execution** for 1,000+ monthly jobs using explicit state transitions, retry handling, and failure isolation, improving processing throughput by **3x** while strengthening reliability under concurrent production workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Engineered **Flask services**, **REST APIs**, and React workflows handling **10,000+ daily requests**, integrating secure file ingestion and automated grading pipelines for customer-facing educational workflows running in production environments.
- Designed **multithreaded batch processing** for 1,000+ uploaded files using AWS RDS-backed services, reducing grading turnaround by **60%** while improving throughput, reliability, and issue diagnosis during peak evaluation periods.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **Python API and ETL microservices** with SQL, AWS Glue, and S3, improving response times by **30%** while supporting resilient data movement, validation, and operational reliability across distributed services.
- Maintained **Jenkins CI pipelines** and 20+ automated test suites with PyTest and Mockito, increasing release reliability by **68%** and strengthening regression control, debugging speed, and deployment confidence across environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **REST API and WebSocket features** supporting **2M+ synchronized records** across modular workflows, improving data consistency and real-time responsiveness for product operations dependent on cross-service coordination patterns.
- Automated **Docker-based build and test workflows** with Jenkins CI/CD, cutting release time by **50%** while improving deployment consistency, troubleshooting speed, and shared validation across staging and developer environments.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*

- Formulated a **quantum-resistant blockchain design** by integrating 5+ lattice-based cryptographic primitives, strengthening decentralized security assumptions and translating research ideas into protocol architecture grounded in distributed consensus concepts.
- Analyzed **blockchain consensus mechanisms** and developed a novel approach reducing centralized-control and quantum-risk exposure by **25%** in benchmark simulations, contributing to a peer-reviewed paper presented at IISc Bengaluru.

Skills

Languages: **TypeScript**, **Python**, JavaScript, SQL, Java, C++, Bash
Backend & APIs: REST APIs, FastAPI, Flask, Node.js, WebSockets, microservices, multithreading, background workers, concurrency
Blockchain & Security: blockchain, consensus algorithms, cryptography, Web3, wallet authentication, decentralized systems, fault isolation, validation checks
Data & Cloud: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, AWS, Azure, S3, Blob Storage, AWS Glue, ETL pipelines, indexing, query tuning
Delivery & Quality: Docker, Jenkins, CI/CD, Git, Kubernetes, PyTest, Mockito, automated testing, reliability engineering, incident debugging, Agile/Scrum

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, Plaid*

- Built a scalable **fintech platform** supporting **50,000+ users** and **1M+ transactions**, combining secure APIs, real-time synchronization, and deployment automation to maintain responsive behavior under high-volume workloads.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting **10+ databases**, accelerating query authoring and schema exploration through a unified Java and TypeScript application with Docker-backed development workflows.

Blockchain Authentication Paper Code | *Blockchain, Authentication, Security Research*

- Designed a **blockchain-authentication model** for three user-access scenarios, replacing centralized verification with a decentralized, tamper-proof login flow and strengthening role-based security assumptions across distributed access patterns.